

Homework 6

35 points

1. Given that $\text{proj}_{\vec{u}} \vec{v} = (3, 6, -3)$ and $\vec{u} = (1, 2, -1)$. Find the possible values of $\vec{v}$
5 points

2. Suppose a vector $\vec{u} = (6, 2, 1)$ that lies in $\mathbb{R}^3$. You also have a space S spanned by $\vec{v} = (1, 1, 0)$ and $\vec{w} = (1, -1, 1)$. **10 + 5 = 15 points**

- Find the projection of $\vec{u}$ on S. Find the co-ordinates in terms of the basis vectors of S.
- Find the component of $\vec{u}$ orthogonal to S

3. A robot arm in 3D moves in the directions spanned by $\vec{v} = (1,0,1)$ and $\vec{w} = (1,1,1)$. If the control force is applied as $\vec{F} = (3,2,5)$, find the component of $\vec{F}$ that lies within the plane of motion. Give the co-ordinates in terms of the basis vectors of the plane of motion. **10 points**

4. Suppose you have the following data:

x	1	2	3
y	2	2	4

Give the line of best fit of the form: $y = mx + c$ **10 points**

5. For $x = (1, 2, 3, 4)$ and $y = (2, 4, 5, 8)$. Find the best fit line through the origin. **5 points**